SHAMIK DEY

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Portfolio | LinkedIn | GitHub | Google Scholar

Professional Experience

Research Assistant Dec 2025 - Present
Ahsanullah University of Science and Technology (AUST)
AUST Internal Research Grant (AIRG): Publicly Accessible Ripe-Unripe Crop Image Dataset & AI-Based Crop Maturity Detection System
Leading end-to-end design of a maturity-classification pipeline tailored to Bangladeshi agricultural crops, from data acquisition and multi-class annotation protocol design through CNN/object-detection model development and evaluation. Structuring the dataset for public release to support reproducible agri-AI research.

Software Engineer Jun 2025 - Present
Sincos Automation Technologies Ltd.
Designed and deployed a voltage anomaly detection and time-series forecasting pipeline for an industrial monitoring application, covering preprocessing, model development, and real-time inference deployment. Concurrently building the frontend architecture for an OEE analytics platform (Angular, Tailwind CSS).

Jr. Software Engineer Jan 2024 - Jun 2025
Sincos Automation Technologies Ltd.
Contributed to OEE software design and a voltage anomaly detection system, gaining early exposure to applied ML deployment alongside full-stack development.

Publications

1. **Shamik Dey**, Adnan Quaium, Jeba Maliha, Md Mehedi Hasan, Md. Abdullah-Al-Wadud, Saifur Rahman Sabuj, and Jia Uddin, "**ResShuffle-ECAFusion: ECA-Guided Dual-Backbone Feature Fusion for Explainable and Uncertainty-Aware Differentiation of Glioma and Meningioma from Brain MRI**", *Scientific Reports Journal (Under Review)*

2. **Shamik Dey**, Adnan Quaium, Saiful Islam, Abdus Shakur Siam, "**EyeDiseaseNet: An Explainable and Resource-Efficient Self-Attention Hybrid Network for Multi-Class Ocular Disease Classification**", *Signal, Image and Video Processing Journal (Under Review)*

3. **Shamik Dey**, Ayon Rahman, Sadique Billah, Adnan Quaium, "**Solar Irradiance Forecasting with a Compact Gradient Boosting Ensemble**", 2025 28th International Conference on Computer and Information Technology (ICCIT) DOI: 10.1109/ICCIT68739.2025.11490233

4. **Shamik Dey**, Adnan Quaium, "**Leakage-Safe Multi-View Graph Construction for Robust Intrusion Detection in Edge-IIoT Networks**," 12th International Conference on Networking, Systems, and Security (NSysS '25). ACM, New York, NY, USA DOI: 10.1145/3777555.3777572

5. Md Ehsanul Haque, Abrar Fahim, **Shamik Dey**, SA Jahan, SM Islam, Sakib Rokoni, MS Morshed, "**A Modified VGG19-Based Framework for Accurate and Interpretable Real-Time Bone Fracture Detection**," 16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT) DOI: 10.48550/arXiv.2508.03739

6. Abrar Fahim, **Shamik Dey**, Md Nurul Absur, Md Kamrul Siam, Md Tahmidul Huque, Jafreen Jafor Godhuli, "**Optimized Approaches to Malware Detection: A Study of Machine Learning**

and Deep Learning Techniques," 2025 IEEE 14th International Conference on Communication Systems and Network Technologies (CSNT) DOI: 10.1109/CSNT64827.2025.10968061

7. SM Mozumder, Tabassum Alamgir Nishat, Bayezid Hasan Bhuiyan, Md Rahabbul Khan, **Shamik Dey**, Adnan Quaium. "**Improving Urban Mobility in Dhaka: Machine Learning-Based Traffic Prediction**," 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) DOI: 10.1109/ECCE64574.2025.11013936

8. FM Mahmudul Haque, Humayra Tabassum, Md Fazal Amin, Md Nazrul Islam Jony, **Shamik Dey**, Adnan Quaium "**Short Paper: Predicting and Analyzing EV Energy Consumption in Bangladesh: A Machine Learning Approach**," 11th International Conference on Networking, Systems, and Security (NSysS '24). ACM, New York, NY, USA DOI: 10.1145/3704522.3704553

Projects

Voltage Anomaly Detection and Prediction System
- Built an end-to-end machine learning pipeline for voltage anomaly detection and time-series forecasting, including data preprocessing, feature engineering, model development, and evaluation.
- Deployed the predictive model for real-time inference, providing automated anomaly alerts and voltage trend prediction for proactive system monitoring.

Production-Ready Multi-Agent AI System with FastAPI
- Engineered a scalable multi-agent AI platform using FastAPI, featuring Retriever, Summarizer, and Orchestrator agents for intelligent information retrieval and response generation.
- Implemented REST APIs, agent orchestration, automated testing (PyTest), and modular deployment using Conda with comprehensive developer documentation.

Adaptive Multi-Agent Mathematics Tutoring System
- Developed an intelligent multi-agent tutoring system that automatically routes mathematical queries to specialized reasoning agents based on educational level.
- Designed modular routing architecture, context-aware reasoning pipelines, and agent-specific explanation strategies for personalized learning experiences.

DayMate: LangGraph-Based AI Personal Planning Assistant
- Developed a LangGraph-powered AI assistant integrating weather services, news retrieval, conversational memory, and reasoning workflows for personalized daily planning.
- Leveraged tool-calling agents, ChromaDB memory, external APIs, and web scraping to generate context-aware recommendations based on user preferences and real-time events.

Education

B.Sc. in Electrical and Electronic Engineering July 2019 - November 2023
Ahsanullah University of Science and Technology
CGPA: 3.293/4.00

Technical Skills & Strengths

Programming Languages:	Python, JavaScript.
Web Technologies:	Angular, React JS, FastAPI, Tailwind CSS, BootStrap, WordPress, PostgreSQL,MongoDB.
Libraries & Frameworks:	Pytorch, TensorFlow, OpenCV, LangChain.
Design & Verification Tools:	UI/UX Design, Cadence Virtuoso, Innovus, Quartus, Proteus, AutoCAD, PSpice, Arduino.
Cloud & General Tools:	AWS, Docker, Git, Google-Suite.

Leadership & Volunteering

Undergraduate Thesis Instructor Feb 2023 - Present
Ahsanullah University of Science and Technology (AUST)
Appointed by Assistant Professor Md. Adnan Quaium to mentor undergraduate final-year students. Providing technical guidance on Machine Learning workflows, dataset selection, and experimental setup. Assisting students in refining their thesis writing and preparing manuscripts for peer-reviewed publication.

Technical Operations, Management & Session Anchor Dec 2023
26th International Conference on Computer and Information Technology (ICCIT)

Treasurer Feb 2023 - Jan 2024
IEEE AUST Student Branch

Assistant Webmaster Mar 2022 - Feb 2023
IEEE AUST Student Branch

Significant Courses

Major Courses in B.Sc.: Computer Architecture, VLSI - I, VLSI - II, Processing and Fabrication Technology, Computer Architecture, Power Electronics, Multimedia Communication.

Professional Certifications: **Machine Learning Specialization** (Stanford / DeepLearning.AI), **Deep Learning Specialization** (DeepLearning.AI), **AI for Medical Diagnosis** (DeepLearning.AI), **Advanced Computer Vision with TensorFlow** (IBM).

Awards

Champion: Undergraduate Thesis Paper Contest at International Student LED Conference
International Student LED Conference 2023 is a co-located conference of IEEE ICCIT 2023 and IEEE ICPS 2023.

Reference

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